

Dhanya Sri Vasantha

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Profile

Highly skilled Data Scientist with a solid foundation in machine learning, statistical analysis, and data cleansing. Experienced in turning complex datasets into meaningful insights, building predictive models and designing scalable data pipelines. Proficient in Python, SQL, and cloud platforms like AWS, with a proven ability to deliver data-driven solutions that drive business growth and improve operational efficiency. Known for effectively communicating technical findings to non-technical stakeholders through clear visualizations and data storytelling. Passionate about using data to solve real-world problems and uncover opportunities for innovation and growth.

Education

George Mason University

MS Data Analytics Engineering, GPA: 3.90/4.0

Fairfax, VA

August 2023 - May 2025

DVR & Dr. HS MIC College of Technology

BTech Computer Science Engineering, GPA: 8.03/10

Andhra Pradesh, India

August 2019 - May 2023

Work Experience

Data Analyst, La Liga de Ciudades de Puerto Rico

(01/2025 - 05/2025 - Arlington, VA)

- Designed and deployed a cloud-based Municipal Data Intelligence Dashboard using Python, MySQL, Streamlit, and AWS EC2, enabling real-time visual insights across 78 regions.
- Built end-to-end ETL pipelines to clean, standardize, and integrate 20+ datasets, improving data quality by 95% and enabling fast, filterable querying through a normalized database schema.
- Collaborated in an Agile, Scrum-based team, using Git for version control and completing 5 sprints to deliver a secure, scalable, and user-friendly platform tailored to real-world decision-making needs.

EduSkills (AI-ML Virtual Internship)

(07/2022 - 09/2022 - AP, India)

- Completed a 10-week AI-ML virtual internship supported by AWS Academy.
- Conducted model evaluations and hyperparameter tuning to enhance model accuracy.
- Processed and analyzed 500,000+ data points to improve model performance by 15%.

Projects

Customer Churn Prediction Using Machine Learning

- Explored 7,000+ telecom customer records to detect churn patterns using data visualization and EDA.
- Built and compared machine learning models; Logistic Regression achieved 80% accuracy and AUC of 0.86.
- Reduced customer churn prediction time by 30% through optimized feature selection and model tuning.

Crime Prediction

- Conducted in-depth analysis of 800K+ crime records from Los Angeles (2020-2023) to examine temporal, categorical, and spatial trends impacted by COVID-19.
- Applied data cleaning, temporal analysis, and spatial mapping using Python, R, and MySQL, integrating NOIR data classification for structured insights.
- Implemented a real-time alert system that reduced crime response time by 20%.

Text Summarization Using NLP

- Built an NLP-based summarization model that processed over 10,000 Wikipedia articles, reducing content length by 70-80% while retaining key information.
- Improved summary coherence by 20% using ROUGE score evaluation metrics.
- Reduced processing time by 35% through optimized text pre-processing techniques.

Technical Skills

Programming & Tools: Python, R, SQL, Jupyter Notebook, Google Colab, VS Code, RStudio, MySQL Workbench, Oracle, Github, Databricks, Jira.

Data Analysis & Visualization: Pandas, NumPy, Tableau, Power BI, scikit-learn, Matplotlib, Seaborn, ggplot, micromapST, MS-Excel plotly, Streamlit.

Machine Learning & NLP: PyTorch, TensorFlow, Keras, Sklearn, NLTK, Predictive Modeling, Regression (Linear, Logistic), Clusterng, Model Evaluation.

Backend & Data Engineering: FastAPI, Flask API, Tkinter, Data Cleaning, Feature Engineering, Data Modeling, ETL Pipelines, Time Series Analysis, Apache Spark, Workflow Automation, ServiceNow.

Certifications

- Microsoft Certified: Azure Data Fundamentals
- Google Data Analytics (Coursera)
- Cloud Foundations, Machine Learning Foundations (AWS)
- Introduction to NLP, SQL for Data Science (Great Learning)
- Introduction to R, Tidyverse & Tidymodels (Data-camp)